

Resilient decentralized communication for unstructured peer-to-peer networks

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Abstract

Recent crises, natural disasters, and large-scale protests emphasize how people rely heavily on online social networks and instant messaging platforms to communicate, while organizations use those platforms to coordinate aid efforts. During crises, communication networks can become strained, partially or wholly destroyed, censored, or otherwise monitored, leaving reliant communities in need of alternatives. One of the issues is that while existing Internet infrastructure is distributed, most of its management and ownership are consolidated. Free distributed alternatives to consolidated social networks are peer-to-peer networks, which distribute the power and content control over many, equal peers. Previous studies evaluated the metrics and performance of structured overlays of p2p networks. This paper evaluates decentralization metrics of unstructured p2p networks, by providing empirical and simulated insights to their performance, resilience, sparsity, and reliability. In this context, we propose design considerations to support the creation of more resilient communication clusters.

1 Introduction

The decentralized administration of the internet is one of the basic enablers of the rapid growth of the Internet as a whole, thanks to the autonomous nature of different network systems. Different Internet Service Providers (ISPs) play a critical role in the functionality of the Internet, as they connect their end customers (Internet users) to the Internet backhaul. The power dynamic of this system inherently gives control over shaping consumption patterns to the ISPs: consolidation of power at the hands of a few ISPs gives them the unique ability to control, throttle, or monitor the flow of information, in addition to enabling oppressive practices such as censorship and surveillance of end users. Most recently, communication disruptions due to airstrikes on main fiber optic lines and radio sites, and the depletion of fuel for generators, have severely limited the information flow to the population of Gaza since the start of the war in October 2023, halting humanitarian coordination, and leaving the affected population entirely cut off from life-saving information and assistance ([?], [?]). In addition to these physical disruptions, governmental control of the telecommunication infrastructure often leads to the repression of communities during times of political unrest ([?], [?]), severely limiting the possibility for social actors to dissent and organize themselves. Together, the aforementioned practices have led to a growing interest in peer-to-peer (p2p) messaging platforms. Most notably, the rise of p2p messaging platforms like Firechat [?], Briar [?], and Amigo [?], have proven that there is a growing shift in user interest patterns away from traditional Centralized Social Networks towards

diverse p2p social networks. Physical p2p networks have a limited communication radius, draw support from their underlying hardware and protocols, and utilize multi-hop communication. There are a plethora of protocols to use for p2p communication, and we chose to focus on stable protocols that most devices support and the populace are most likely to use in a disaster scenario: virtually all smartphones support Bluetooth and smartphone usage is largely ubiquitous, even in refugee camps and disaster zones. By focusing on low-power communication, device battery life can be conserved to allow for critical communication in extreme situations: we thus decided to use Bluetooth Low Energy (BLE) for our p2p communication protocol, which allows users to exchange messages over short ranges. To achieve a higher level of information decentralization, and resilience to topology changes, we decided to focus on unstructured p2p networks as they are more suitable for dynamic alternative communication methods often necessary to guarantee free speech in disaster zones, during protests, and for citizens living under oppressive regimes. The research community has proposed many metrics to measure the centralization, sparsity, cost, and resilience of global ISPs, Content Delivery Networks (CDNs), Certificate Authorities (CAs), blockchains, and signals ([?], [?], [?]), algorithms for selecting the most appropriate communication mode in Bluetooth 5 [?], and different signal propagation path loss models [?]. Most recently, systems and protocols were proposed to enable secure p2p communication between smartphones ([?], [?], [?]), but none of those works, to our knowledge, evaluate the metrics that prove those systems are actually resilient and appropriate to use in extreme real-world situations. In this research paper, we conduct a comprehensive representational experiment to collect empirical data on the stochastics of the BLE signal propagation, and use the collected real-world empirical evidence to parameterize simulations of unstructured p2p networks, and then measure their centralization [?], sparsity [?], and theoretical channel capacity, to elicitate a claim about the resilience of local networks.

Contributions. To summarize, this paper makes the following contributions:

- We present results of a comprehensive experiment to collect empirical data on BLE signal propagation stochastics conducted across different distances between transmitter and receiver.
- We show how the collected empirical evidence could be used to parameterize simulations of message passing in unstructured p2p networks at scale based on Bluetooth Low Energy.

- To evaluate unstructured p2p networks, we capture and present throughput, latency, centralization, and sparsity metrics measured in network simulations.
- We show examples for design considerations for p2p overlays that outperform state-of-the-art overlay designs based on throughput and communication latency.

In section ?? we discuss the Bluetooth protocol background, the state of the art of sparsity and resilience metrics as measurable in network topologies, and present related works. Section ?? describes the setup of the experiment testbed that captures the BLE measurements, presents the empirical results from the testbed which we then use as parameters for the simulations, and closes with detailing the simulation setup ?? . Following the setup section, section ?? evaluates the results from the simulations, concluded by the discussion on section ?? .

2 State of the Art

In disaster zones, for example after natural disasters or wars, networking infrastructure is oftentimes destroyed or targeted. In sparsely populated areas of the world, such infrastructure might not be available at all: the Lapponia region near the Arctic Circle, homeland of the Sámi people, is characterized as a sparsely-populated region full of icy fjords and mountains. To provide decentralized p2p communication, Lindgren et al. move away from traditional IP packet forwarding and propose the Store-Carry-Forward paradigm using Delay-Tolerant Networking (DTN) [?], where hosts store messages while carrying them until they rendezvous with another node to forward them. This paradigm enables the opportunistic usage of technologies such as Wi-Fi Direct or Bluetooth to forward messages, which is ubiquitous to virtually all Smartphones and computers today. Decentralized Social Networks (DSNs) can exist both over standard IP packet routing, or as DTN-based communication platforms supplemented by the Internet insofar as it becomes available, as Legendre et al. present with Twimight [?]. Regarding consolidation of services, DSNs can function as feudal-type subsystems, where users are given the ability to run the entire software stack as their own server, uniting around a set of rules or constitution (cf. Mastodon [?]), or as a collection of transparent community-operated competing components interoperating on the same server, like in Bluesky [?]. Structured peer-to-peer networks such as Chord [?], Kademlia [?], and the InterPlanetary File System (IPFS) [?] use Distributed Hash Tables (DHTs) to store and retrieve (meta-) content in a decentralized manner. On the other hand, unstructured peer-to-peer networks are more dynamic and do not use DHTs to spread centralized information. Instead, unstructured peer-to-peer networks rely on flooding, efficient clique-level forwarding, and gossip for content dissemination, as recently exemplified with Amigo [?] and bitchat [?], as they do not store information about the cluster membership. The concepts of flooding and clique-forwarding are the core principles of this research paper, as the resilience and performance metrics have been measured as factors of decentralization of unstructured peer-to-peer social networks, that are built using flooding and clique-forwarding for neighbor discovery and content dissemination. Different countries have wildly varying centralization levels of their web service

providers, which directly impacts their resilience to outages, content delivery latency for end-customers, the cost of communication, and the relative freedom of Internet usage. In [?] Habib et al. study patterns of web service hosting in different countries, and demonstrating how countries with higher centralization around a smaller number of website hosting providers are more susceptible to outages, and propose a metric called the Centralization Score $\mathcal{S}$, which Habib uses to measure the centralization of website hosting around a small number of providers, DNS service providers, and CAs. The Centralization Score $\mathcal{S}$ is derived from the Herfindahl-Hirschmann Index (HHI), which is used to measure market competition in US antitrust laws. To use the different sparsity metrics we will introduce, the leading metric I used to measure centralization and sparsity is the node degree: the number of connections a node has to other peers in the network topology. An overall similarity between individual node degrees increases the resilience of a topology, as a network partition affecting a subset of the nodes will not result in a complete blackout of the network because many alternative paths still exist. The Centralization Score $\mathcal{S}$ is defined as follows:

$$\mathcal{S} = \sum_{i=0}^n \left(\frac{a_i}{C} \right)^2 - \frac{1}{C}$$

where C is the total number of websites considered, a_i is the number of websites hosted by provider i , and n is the total number of providers. In this paper, C is defined as the total number of links between nodes, a_i is the node degree of node i , and n is the total number of nodes. Formally, let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph, where $\mathcal{V}$ is the set of nodes (vertices), and $\mathcal{E}$ is the set of links (edges). If $(i, j) \in \mathcal{E}$ then $(j, i) \in \mathcal{E}$, i.e., links are bidirectional. There can only exist one link between every pair of nodes, a link increases the node degree of both participating nodes by one, and nodes are not linked to themselves (the node degree of an isolated node, i.e. a node that has no links, is defined as zero). To sparsity: in their work, Hurley and Rickard compare different measures of sparsity [?], and rank different known sparsity measures according to six desirable criteria that a sparsity measure should trivially satisfy, among them: the Robin Hood axiom ("stealing from the rich and giving to the poor", or in the context of distributed networking: website hosting over multiple providers instead of a single provider should reduce sparsity); the Scaling axiom (sparsity is scale-invariant); and the Rising Tide axiom (sparsity is decreased by adding a constant to all coefficients). Hurley et al. prove that amongst the most common methods to measure sparsity, the Gini coefficient outperformed all other measurements, and satisfies all six criteria. The Gini coefficient, a metric commonly associated with the Human Development Index to measure income inequality around the world, is calculated as follows:

$$G = \frac{\sum_{i=0}^n \sum_{j=0}^n |a_i - a_j|}{2n \sum_{i=0}^n a_i}$$

Where a_i is the amount of resources of node i , and n is the total number of nodes. In this paper, a_i is defined as the node degree of node i . Hurley and Rickard's work allows us to effectively use tried-and-tested economical metrics to measure the sparsity of the different simulation topologies. To conclude, this paper compares the aforementioned metrics with the trivial L0 norm, which is

calculated from the different topologies' node degrees followingly:

$$L_0 = \sum_{i=0}^n \begin{cases} 1 & \text{if } a_i > 0 \\ 0 & \text{otherwise} \end{cases}$$

where a_i is the node degree of node i . To summarise, this paper uses the Centralization Score $\mathcal{S}$ and the Gini Coefficient G , and compares them with a basic metric, the L0 norm, to evaluate different topologies and rank them according to their centralization and sparsity metrics.

3 Setup

3.1 Testbed

Initially contemplating measuring WiFi signal propagation as well, I decided to focus on Bluetooth as it is more power efficient, and requires no network infrastructure. To measure the BLE signal coverage stochastics and how it propagates in free space, the Received Signal Strength Indicator (RSSI) of BLE advertisements at the receiver node is compared across increasing distances from the transmitter node. Furthermore, Bluetooth Low Energy (BLE) was picked over classic Basic Rate/Enhanced Data Rate (BR/EDR), as it is even more efficient for (small) message exchanges [?]. BLE advertisements form the bottleneck of BLE communications, as ads are used to discover and connect to neighbours in local networks. To discover peers, peers have to scan for BLE advertisements and not advertise at the same time, due to the half-duplex nature of BLE. Two types of scanning are possible: passive scanning, where the scanner only listens for ads, and active scanning, where upon receiving an ad, the scanner also sends extended scan requests to the advertiser node for additional data. In this experiment, passive scanning is used, because it is sufficient to discover neighbors and characterise the RSSI.

Our testbed consists of a set of Raspberry Pi 3B+ devices, using the built-in BLE supporting chip Cypress CYW43455, and external power supplies of 5v/3.4A to ensure stable operation and mobility. The Raspberry Pis were flashed with a lightweight 64-bit Linux port (Raspberry Pi OS Lite based on Debian Trixie), upon which the basic Bluetooth stack was installed using BlueZ 5.66.

3.2 Empirical Insights

During the experiment, one Raspberry Pi acted as the transmitting node, using an automated bluetoothctl script to continuously emit BLE advertisements at a fixed interval of $20\text{ms} + X$, where X is a random delay uniformly distributed between 0 and 10ms to avoid collisions with other devices using the same advertising interval ([?], [?]). Another Raspberry Pi performed the receiver role, running a Python script that uses the Bleak library to communicate with the underlying BlueZ BLE stack to scan for a window set to one minute, during which the receiver peer continuously scanned for advertisements from the surrounding environment. The two devices were placed at increasing distances from each other, where the advertiser node was kept static and the receiver node was moved. At every distance step, the receiver node recorded the RSSI of all ads received from the transmitting node, filtering them by the unique address of the advertising node, and logging the RSSI values, as well as the transmitting power (tx) and a timestamp, to a file. Each distance step was repeated twenty times to capture enough data

points for statistical significance. The distances measured were from 0 meters to 10 meters in increments of 5 meter, and then from 10 meters in increments of 10 meters, until no more ads were received for most runs of that distance. The experiment was divided to two parts: in the first part, the distance points up to 30 meters were evaluated in an urban backyard; in the second part, the distance points from 40 meters were evaluated in an olympic athletics field. Both parts of the experiment were conducted in an open level environment with minimal surrounding interference, clear direct line of sight between the nodes, and minimal multipath effects.

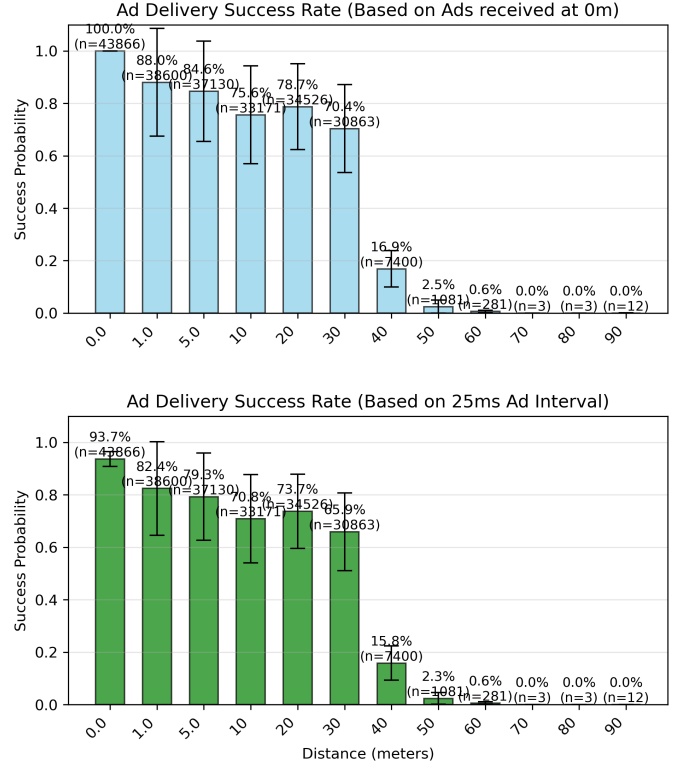


Figure 1: Advertisement Delivery Success Probability

Figure ?? shows the advertisement delivery success probability at different distances, or the percentage of ads received at the receiver node, depending on two estimates of the number of ads transmitted. The figure showcases two subplots: the top plot calculates the ratio between the count of ads received at distance X , and the count of ads received at 0 meters, where the two devices were placed right next to each other. This plot assumes that the count of ads received at 0 meters is the same as the count of ads that were actually sent, and takes it as a baseline. The bottom plot estimates the number of ads sent by the transmitter node by dividing the length of time of the scanning interval by the known advertising interval on the advertising node of 20ms and adds to it the expected value of the random variable representing the delay introduced by the underlying hardware: 5ms. It is trivial to see that as the distance increases, the advertisement delivery success probability decreases, following a roughly exponential decay. No ads were received for

any run for distances more than 100 meters, so the experiment was concluded at that point.

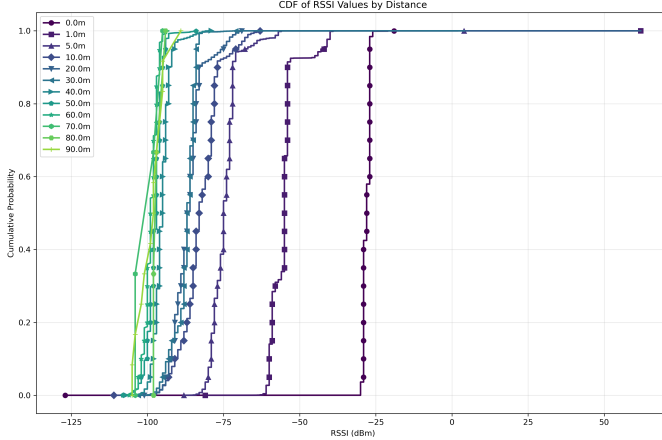


Figure 2: RSSI CDF

The Cumulative Distribution Function (CDF) plot of the RSSI values received at different distances is shown in figure ???. As expected, the figure shows a general decreasing trend of the RSSI values, in correlation with the increase in distance. The jittery nature of the CDF curves at distances ≥ 40 meters can be attributed to the lower number of samples collected at those ranges, which leads to a higher variability in the RSSI measurements. The Probability Density Function (PDF) plot and a time series plot of the RSSI values for each distance can be found in the appendix, figures ??? and ???. Next, table ?? shows summary statistics of the RSSI values received

Table 1: RSSI Statistics by Distance

Distance (m)	Mean RSSI (dBm)	Std Dev	Num Samples
0.0	-28.03	1.21	43,866
1.0	-55.14	4.62	38,600
5.0	-74.58	4.77	37,130
10.0	-82.47	6.21	33,171
20.0	-86.54	5.05	34,526
30.0	-87.45	3.55	30,863
40.0	-95.21	2.76	7,400
50.0	-97.50	2.44	1,081
60.0	-98.82	2.10	281
70.0	-99.00	3.74	3
80.0	-96.33	1.70	3
90.0	-98.25	4.23	12

at different distances: the mean RSSI values presented at the table were used to parameterize the simulation signal propagation decay with distance. Throughout all distances, the same transmitting power was recorded which hints at the default transmitting power of a Raspberry Pi 3b+: 12 dBm.

3.3 Simulations

We used the The ONE simulator [?] to simulate an opportunistic exchange of messages between nodes communicating over the BLE, using the empirical results from the testbed experiment to model the signal path loss with distance. The simulation was repeated for 25 runs per configuration, using an Ubuntu 22.04 with a Intel(R) Xeon(R) Gold 6326 CPU @ 2.90GHz with 64 cores, and 247 GB of RAM. To run the simulations we used parallelization by spawning 60 parallel jobs. In each run, the nodes remain static and attempt to send the same amount of messages to all other nodes in the network, using an epidemic router to flood messages across the network. The epidemic router is especially useful to simulate the scenario of message propagation in disaster zones, where it is critical to disseminate information as fast and as wide as possible. In an epidemic routing scheme, every node sends all its messages to all of its neighbors, and each neighbor recursively repeats this iterative process until the final destination node for each message is reached. In every simulator run, a host generates 20 messages to all other hosts in the network, with a size of 247 bytes each. A simulator run ends when all generated messages are either delivered or discarded.

For the signal path loss model, we used the empirical data presented in table ??, and interpolated for distances between the empirical measurements using the normal logarithmic path loss model laid out in equation 6 of [?]. The path loss model is defined as follows:

$$RSSI_d(dBm) = A - 10 * \alpha * \log_{10}\left(\frac{d}{d_0}\right)$$

where $RSSI_d$ is the received signal strength indicator at distance d between source and destination nodes, d_0 is the distance at the nearest lower or equal reference point, A is the RSSI at the nearest lower or equal reference distance d_0 (as measured by the testbed), α is the path loss exponent. We wanted to evaluate the performance of the Bluetooth Low Energy running on 1M PHY mode, which means a symbol modulation rate of 1Mbps at ≤ 1 meter.

A large 400×400 meter L-shape building hall was generated, split to a grid of 12 equal-sized cells of 100×100 meters each assigned to a cluster consisting of six hosts, for a total of 72 hosts. Each host was randomly placed within the cluster cell assigned to it. A random number generator that generates the node positions, whose seed is derived from the communication range, the message size, and optionally by the simulation run number, is used to ensure reproducibility of results. The communication range is a configurable parameter which limits the radius of communication of the network interface that each node has: for simplification, a connection between two nodes can be established if they are within the communication radius of each other, and have direct line of sight. A connection does not churn during a simulation run: once a connection is established it is maintained until the end of the simulation run.

3.4 Model Insights

Different topologies were generated by using the communication radii of [1,5,10,20,30,40,50,60,70,80,90,100] meters. A simulation run configuration consists of a set of parameters, including the message generation rate, the maximum communication radius of each

node, the maximum number of parallel connections (max node degree) each node can establish and maintain, and the communication mode: either exclusively intra-cluster communication to simulate clique-based short-range cluster communication, or inter-cluster communication to simulate long-range communication. The same exact topology was repeated for [1,2,3,4,5,6,7,8,9,10,15,20] maximum node degrees, each for 25 runs, keeping the topologies invariant to the run number but changing them for each communication range, thereby shuffling the order of the hosts picked for message generation, to prevent introducing bias by the alphanumerical ordering of the hosts IDs. This was done to elicit the effects of the maximum node degree and communication range on timely message delivery.

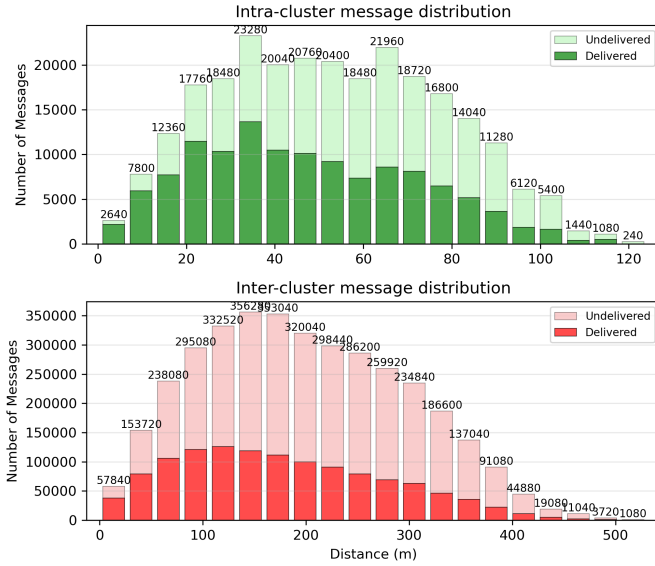


Figure 3: Message Distance Distribution

Figure ?? shows the distance distribution of all messages created, aggregated over all simulation runs, communication ranges, and maximum node degrees tested, split by the communication mode. It is evident to see that the intra-cluster communication mode had a generally higher overall delivery probability, possibly attributed to a lesser strain on the individual hosts' message buffers. To better

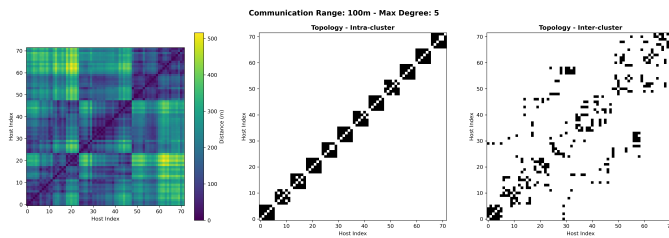


Figure 4: Topology and distance metrics for communication radius 100m, constrained maximum node degree 5

understand the topology, figure ?? shows the distance matrix between hosts to the left, supplemented by two topological matrices,

one for each communication mode. The topological matrices have a black cell at position (i, j) if host i is connected to host j .

Next, randomized topologies were created for each configuration, to assess how the maximum node degree affects the message latency, centralization, and sparsity metrics of unstructured p2p networks. This was done to evaluate how different topologies with the same communication radius and maximum node degree induce variance in the different metrics

4 Results

The figure ?? shows how the latency of message delivery per communication radius changes with an increase in the maximum constrained node degree. The different communication radii are shown as distinct curves on the plot, and shows as expected a trend towards a lower latency with a lower communication radius. A lower communication radius in turn creates fewer messages, as there are generally fewer links and peers to propagate messages to, which exhibits less strain on the message buffers of the individual hosts. An important insight that I derive from this plot is that increasing the maximum node degree beyond a certain point does not significantly improve the latency of message delivery, and that increase is echoed in longer communication ranges: in other words, the longer the communication range, the less effect an increase in maximum node degree has on the latency of message delivery. Surprisingly, higher communication ranges exhibit a positive effect on latency with an increase in maximum node degree, as especially visible in the 100m communication range.

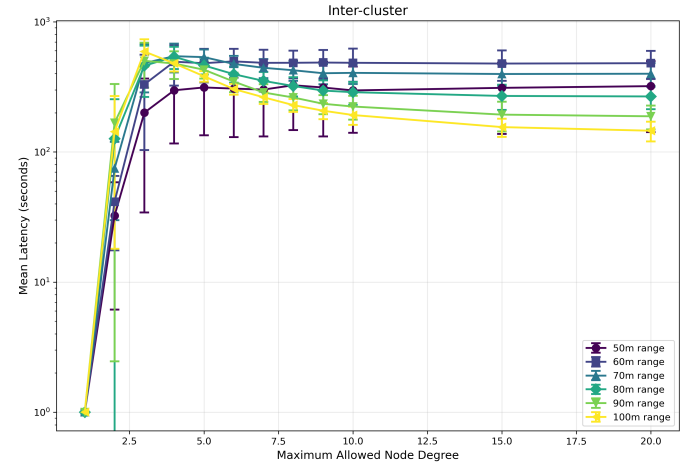


Figure 5: Latency vs. max node degree - inter-cluster communication

Figure ?? shows how the different centralization metrics introduced in section ?? correlate with an increase in the maximum constrained node degree. An extended version of this plot, comparing the aforementioned centralization metrics to the L0 norm also in the intra-cluster communication mode, can be found in the appendix, figure ??.

To better understand how fast messages are delivered in different topologies, we generated successful delivery under time

Inter-cluster Centralization Metrics vs Maximum Node Degree

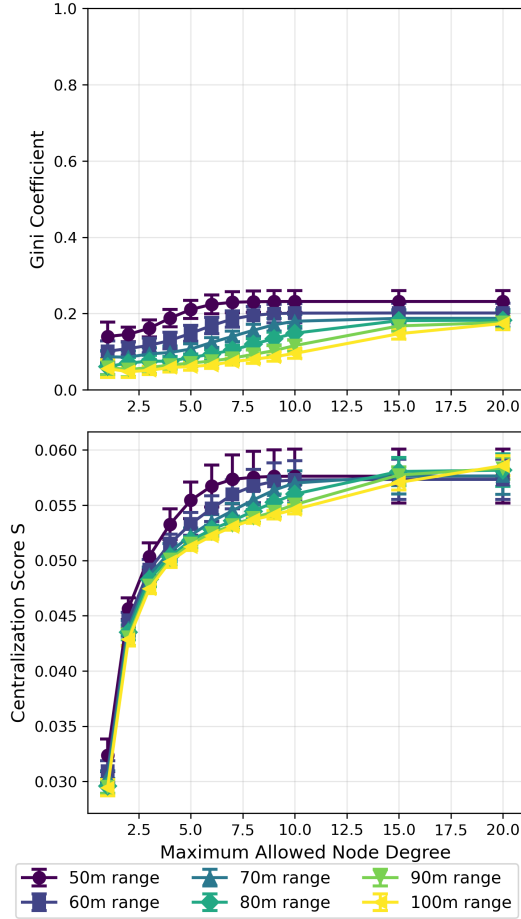


Figure 6: Centralization vs. max node degree - inter-cluster communication

threshold heatmaps for different communication ranges and max node degrees, which provide the full image together with the topological matrices introduced in subsection ??, as to how messages are exchanged in a specific topology. The heat maps that match the example in figure ?? shows two identical topologies, in which the communication range is set to 100m, and the maximum node degree is arbitrarily set to 5. The matrices have hosts along their X and Y axes, and a cell at position (i, j) is colored more green the more messages host i successfully delivers to host j within the configured time threshold. The difference between the two matrices is the time threshold used: for the left plot a tight threshold of 10 seconds was used, while a more relaxed 60s threshold was used for the latter; as expected, the delivery probability improved by about 3x from 4.7% to 14.2%.

5 Evaluation

Currently, due to time constraints, some RSSI values are missing from the dataset, and some must be repeated, as evident from figures

figures/range100_maxdeg5_inter_thresholds_10s_60s.png

Figure 7: Delivery success probability under time thresholds, 100m range, max node degree 5, inter-cluster communication

?? and ??: the 50 and 100 meter plots are missing, the 80, 90, and 110 meter plots are jittery - some of which due to having an abnormally low received packet count, which indicates the experiment should be repeated in future work. The RSSI values we have provided for the ranges that are complete and stable, are mostly sufficient to derive the necessary path loss parameters for the simulation, as they exhibit the expected power-law distribution and are mostly consistent. Figure ?? interestingly shows that latency increases with distance, which confirms the path-loss model, but exhibits an unexpected result: we expected the 120m radius simulation result to have the highest latency on average, as it created the most messages and suffers the most from the epidemic routing's flooding of messages, but it was dominated by 50-70m ranges, perhaps due to the missing RSSI values at those distances. The centralization metrics plot (Figure ??) summarizes the simulation results, and proves that L0 loss is indeed not enough to characterise centralization score: it has 1.0 for most values. The Gini index and the Centralization score S plots show a positive relation: as the Gini index increases, the S score increases as well, but the curves that dominate the graph are the not the same: the Gini index is dominated by the 40m range, while the S score is dominated by the 120m range.

6 Discussion

a point for future research could be to measure the maximum achieved distance with varying transmitting power levels using a different chip or a wireless dongle. Furthermore, we were not able to have fine-grain control over exactly when the advertisements were sent, because the BlueZ Bluetooth communication stack and the

underlying hardware controller handle the advertisement scheduling internally to prevent antenna overheating. To best mitigate this, we set the advertisement interval window to the minimum possible values of to achieve the highest consistent frequency of advertisements sent. We were however able to prove that the advertisements were not actually being consistently sent with the configured interval window of 20ms using a Bluetooth sniffer on a mobile phone, which measured the actual advertisement intervals: these fluctuated around 20-100ms. Another point for future research would be to develop a Bluetooth driver that allows fine-grain control over the advertising scheduling; for that the `python-dbus` library could

be used to directly interface with the underlying communication data bus, D-Bus. We can also here observe a sudden unexpected drop in the number of advertisements received (for example, at 20 meters), which we again attribute to temporary environmental interference (e.g. runners at the track passing by the experiment obstructing the clear path, sometimes with phones or other metal objects in their pockets that reflect the signals and cause more wireless interference).

7 Appendix

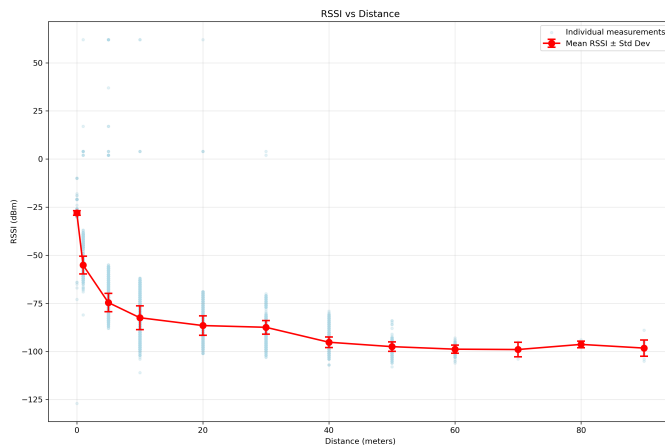


Figure 10: Mean RSSI at different distances

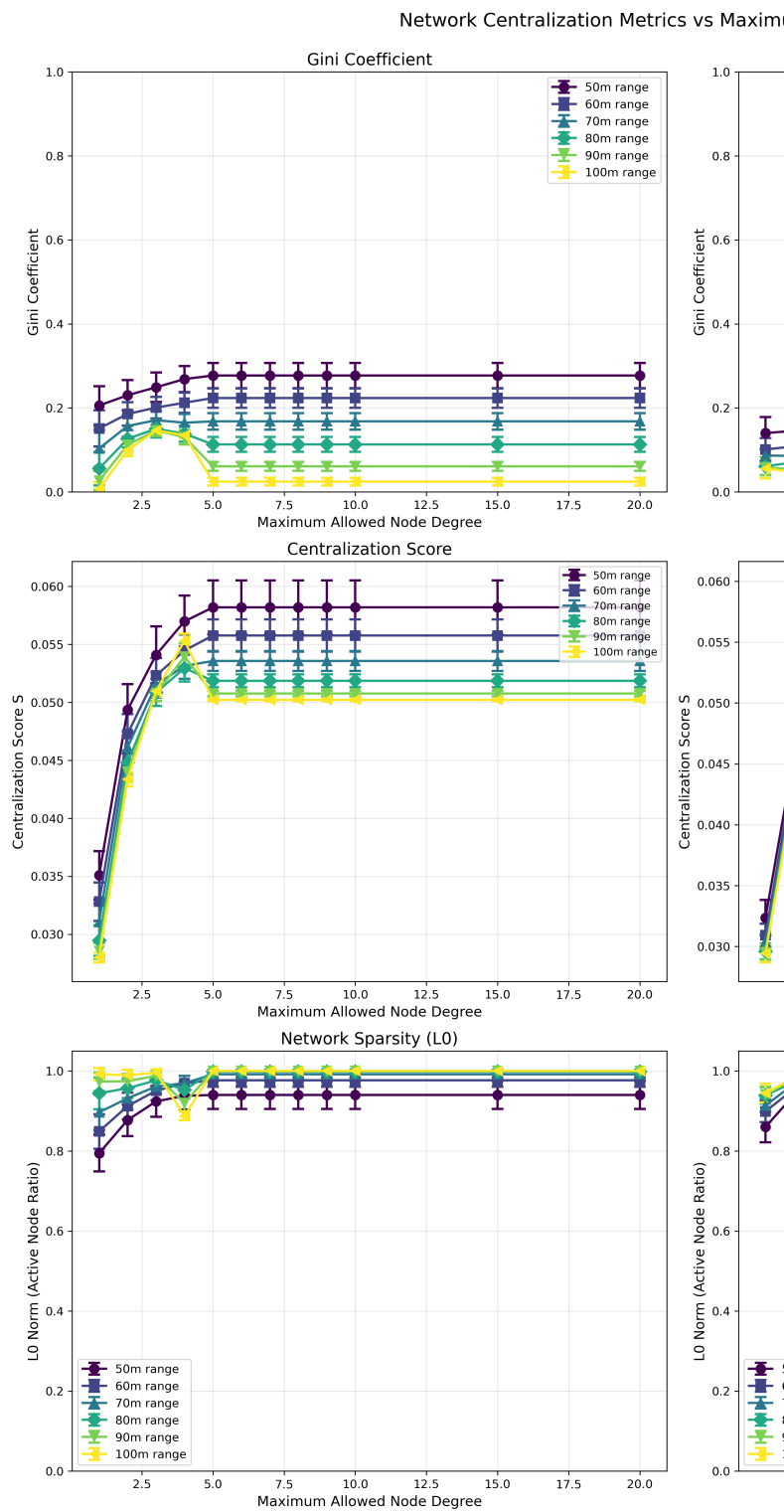


Figure 11: Centralization metrics vs maximum node degree

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